

When Composition Emerges: A Complexity-Ordering Account of Shortcut Preference and Compositional Transience in Transformers

Notes for Rathin

Abstract

Two phenomena in Transformer generalization sit uneasily together in the literature: Wurgaft et al.’s *transient generalization* — in-context generalization that gives way to memorization at high training time — and Khandelwal & Pavlick’s *shortcut preference* — linear composed tasks bypass compositional processing entirely. We give a unified Bayesian rational-analysis account in which both phenomena fall out of a single quantity: the complexity gap $\Delta K = K(G) - K(C_{GG})$ between the atomic and factored descriptions of the task family. The framework predicts three phase diagrams indexed by $\text{sign}(\Delta K)$: for $\Delta K > 0$ (bottleneck tasks), compositional generalization emerges at high D and moderate N ; for $\Delta K < 0$ (linear tasks), the compositional region disappears entirely and only Wurgaft’s two-region diagram remains; for $\Delta K \approx 0$ (marginal tasks), the compositional region is thin and easily suppressed. We derive the closed-form phase boundaries, prove the ΔK -monotonicity in the bottleneck size $|\mathcal{Z}|$, and give a concrete experimental program: sweeping $|\mathcal{Z}|$ at fixed $|\mathcal{X}|, |\mathcal{Y}|$ walks ΔK through zero and should morph the empirical phase diagram from three-region to two-region at a predicted critical value $|\mathcal{Z}|^*$.

1 Introduction

Two robust empirical phenomena constrain any account of compositional generalization in Transformers.

Wurgaft et al. (2025) study in-context learning of atomic tasks and find *transient generalization*: at moderate training time N and moderate task diversity D , the model generalizes to unseen tasks; at high N , it reverts to memorizing the training tasks. They derive a closed-form Bayesian model in which the log-posterior odds between a memorizing predictor M and a generalizing predictor G crosses zero at $N_{MG}^*(D) \sim D^{\beta/(1-\alpha)}$.

Khandelwal & Pavlick (2024) study compositional tasks $h = f \circ g$ and find *shortcut preference*: when the composite admits a linear input-output map, LLMs learn the map directly and show no signature of the intermediate variable $f(x)$ in the residual stream, regardless of training regime. Their finding: the choice of mechanism is mediated by the embedding geometry — when a linear shortcut exists, the model takes it.

These findings sit uneasily together. Wurgaft’s diagram suggests generalization is a function of (N, D) : with enough of both, the model generalizes. Khandelwal–Pavlick’s finding suggests (N, D) cannot force composition when a shortcut exists. What determines when composition emerges, and what determines when even indefinite training cannot force it?

We give a unified account. The framework organizes both phenomena around the *complexity gap*:

$$\Delta K := K(G) - K(C_{GG}),$$

the description-length difference between the atomic and factored priors over composite tasks. The framework predicts:

- When $\Delta K > 0$ (bottleneck tasks), compositional generalization C_{GG} emerges at high D and moderate N , resisting Wurgaft-style transience longer than monolithic generalization G would.
- When $\Delta K < 0$ (linear tasks with cheap atomic descriptions), the C_{GG} region disappears from the phase diagram at every (N, D) . Only Wurgaft’s two-region diagram $\{G, M\}$ remains. This is Khandelwal–Pavlick’s shortcut regime.
- When $\Delta K \approx 0$ (marginal tasks), the compositional region is thin, requires large D to appear, and is easily suppressed by finite-sample loss noise.

The framework thus makes a two-stage prediction. *Whether* composition can emerge is determined by ΔK — a property of the task family, not of training. *When* it emerges in (N, D) is determined by the phase boundaries within the appropriate regime.

Contributions.

- A six-predictor hypothesis space extending Wurgaft’s $\{M, G\}$ to compositional tasks: $\{M, G, C_{MM}, C_{MG}, C_{GM}, C_{GG}\}$ organized by atomic-vs-factored prior and per-stage memorize-vs-generalize choice.
- A three-factor posterior $P(Q \mid S_{T_{\text{train}}}) \propto 2^{-K(Q)^\beta} \cdot \Psi(Q, \theta_N) \cdot \exp(-N_{\text{eff}} \ell_Q)$ that adds an architectural-feasibility factor Ψ to Wurgaft’s complexity-prior–likelihood form, capturing whether the Transformer can implement a given predictor.
- The complexity gap ΔK as the central axis of the framework, with three phase diagrams indexed by $\text{sign}(\Delta K)$.
- A monotonicity result: for random lookup-table composition with $|\mathcal{X}| = |\mathcal{Y}| = m$ and bottleneck $|\mathcal{Z}| = k$, ΔK is monotone decreasing in k and crosses zero at $k^* = m/2$.
- A concrete experimental program: sweeping k at fixed m walks ΔK through zero and should morph the empirical phase diagram from three-region to two-region at $k = k^*$.

2 Setup and Notation

2.1 Compositional task family

Each task is a composite function $h = f_{w_f} \circ g_{w_g}$ with $g_{w_g} : \mathcal{X} \rightarrow \mathcal{Z}$ and $f_{w_f} : \mathcal{Z} \rightarrow \mathcal{Y}$, parameterized by stage parameters (w_g, w_f) . The true generative process is $T_{\text{true}} = T_g \otimes T_f$: each task draws $w_g \sim T_g$ and $w_f \sim T_f$ independently. The training distribution T_{train} is the discrete uniform over D specific pairs $\{(w_g^{(d)}, w_f^{(d)})\}_{d=1}^D$ sampled from T_{true} at the start of training.

2.2 Notation table

Training and posterior	
N, D	Training steps; task diversity
$N_{\text{eff}} = \gamma N^{1-\alpha}$	Effective examples processed by step N
$\alpha, \beta, \gamma, \tau_R$	Framework parameters (four total)
θ_N	Network parameters at step N
Predictors	
M, G	Atomic: memorizing / generalizing over composites
C_{XY}	Factored: g -strategy $X \in \{M, G\}$, f -strategy $Y \in \{M, G\}$
$\mathcal{H}$	$\{M, G, C_{MM}, C_{MG}, C_{GM}, C_{GG}\}$
Posterior ingredients	
$K(Q)$	Description-length complexity
ℓ_Q	Per-example cross-entropy of Q on $S_{T_{\text{train}}}$ (Wurgaft’s $\bar{L}_Q$)
$\Psi(Q, \theta_N)$	Architectural-feasibility factor
η_{Q_1, Q_2}	Log-posterior odds
$\Delta K = K(G) - K(C_{GG})$	<i>Complexity gap: the central quantity of the framework</i>
Routing	
$\rho(\theta) \in \{0, 1\}$	Indicator: routing circuit present in θ
$R(N) = P(\rho(\theta_N) = 1)$	Probability circuit formed by step N
τ_R	Inverse rate of circuit discovery
Δ_{rout}	Loss penalty when circuit absent
Stage structure	
T_g^M, T_g^G	Discrete / continuous priors on w_g
k_g, k_f	Per-stage description costs (one primitive)
K_g^G, K_f^G	Description costs of the true stage distributions
c	Cost of the composition operator

2.3 Wurgaft baseline

For atomic tasks and hypothesis space $\{M, G\}$, Wurgaft et al. derive

$$\eta_{MG}(N, D) = \gamma N^{1-\alpha}(\ell_G - \ell_M) - \log 2 \cdot (K_M(D)^\beta - K_G^\beta), \quad N_{MG}^*(D) \sim D^{\beta/(1-\alpha)}.$$

3 Six Predictors and the Three-Factor Posterior

3.1 Six Bayesian predictors

Two structural choices define a compositional predictor: atomic vs. factored prior, and memorize vs. generalize per stage.

	Atomic prior on w_h	Factored prior on (w_g, w_f)
Memorize both stages	M	C_{MM}
Memorize g , generalize f	—	C_{MG}
Generalize g , memorize f	—	C_{GM}
Generalize both stages	G	C_{GG}

$$M(s_i \mid s_{1:i-1}) = \mathbb{E}_{(w_g, w_f) \sim T_{\text{train}}}[(f_{w_f} \circ g_{w_g})(s_i \mid s_{1:i-1})], \quad (1)$$

$$G(s_i \mid s_{1:i-1}) = \mathbb{E}_{w_h \sim T_{\text{true}}^{\text{joint}}}[h_{w_h}(s_i \mid s_{1:i-1})], \quad (2)$$

$$C_{XY}(s_i \mid s_{1:i-1}) = \mathbb{E}_{w_g \sim T_g^X, w_f \sim T_f^Y}[(f_{w_f} \circ g_{w_g})(s_i \mid s_{1:i-1})]. \quad (3)$$

Assumption 1 (Genuine factorization). $T_{\text{true}} = T_g \otimes T_f$.

3.2 Architectural feasibility

Compositional predictors require the Transformer to propagate g 's output to f 's input within a single forward pass — a routing circuit.

Definition 2 (Routing latent). $\rho : \Theta \rightarrow \{0, 1\}$, with $\rho(\theta) = 1$ iff θ contains a functioning routing circuit (operationalized via bridge-layer activation patching restoring compositional behavior).

Definition 3 (Routing probability). $R(N) := P(\rho(\theta_N) = 1)$, taken over training randomness.

Assumption 4 (Formation curve). $R(N) = 1 - e^{-N/\tau_R}$.

Definition 5 (Architectural-feasibility factor).

$$\Psi(Q, \theta_N) = \begin{cases} 1 & Q \in \{M, G\}, \\ \exp(-N_{\text{eff}}(1 - R(N))\Delta_{\text{rout}}) & Q \in \{C_{MM}, C_{MG}, C_{GM}, C_{GG}\}. \end{cases}$$

3.3 Three-factor posterior

$$P(Q \mid S_{T_{\text{train}}}) \propto \underbrace{2^{-K(Q)^\beta}}_{\text{complexity prior}} \cdot \underbrace{\Psi(Q, \theta_N)}_{\text{architectural feasibility}} \cdot \underbrace{\exp(-N_{\text{eff}}\ell_Q)}_{\text{Bayesian likelihood}}. \quad (4)$$

The pairwise log-odds decomposes cleanly:

$$\eta_{Q_1, Q_2}(N, D) = \underbrace{N_{\text{eff}}(\ell_{Q_2} - \ell_{Q_1})}_{\text{likelihood ratio}} + \underbrace{\log \frac{\Psi(Q_1)}{\Psi(Q_2)}}_{\text{feasibility ratio}} + \underbrace{\log 2 \cdot (K_{Q_2}^\beta - K_{Q_1}^\beta)}_{\text{prior ratio}}. \quad (5)$$

The feasibility ratio is 0 when both predictors are atomic or both compositional; it is $\pm N_{\text{eff}}(1 - R(N))\Delta_{\text{rout}}$ only for atomic-vs-compositional comparisons.

4 The Complexity Gap ΔK

The central object of the framework is the complexity gap between the atomic and factored generalizing predictors:

$$\Delta K := K(G) - K(C_{GG}).$$

Its sign determines whether the C_{GG} region exists in the phase diagram at all.

4.1 Why ΔK is the central quantity

Consider the G -vs- C_{GG} pairwise odds. Both are generalizing predictors with $\ell_G \approx \ell_{C_{GG}}$ on ID data (both converge to ℓ^*). Post-routing ($R(N) \approx 1$), both have $\Psi = 1$ up to an $O(1)$ correction. So Eq. (5) simplifies to

$$\eta_{C_{GG},G} \approx \log 2 \cdot (K_G^\beta - K_{C_{GG}}^\beta).$$

The sign of $\eta_{C_{GG},G}$ is the sign of ΔK : when $\Delta K > 0$, C_{GG} beats G on the prior and the C_{GG} region exists; when $\Delta K < 0$, G beats C_{GG} everywhere and the C_{GG} region is empty.

No finite (N, D) can flip this. The likelihood term is ≈ 0 and the feasibility term is ≤ 0 for C_{GG} . So the framework's central claim is that ΔK is a task-family property, evaluated once, that determines which phase diagram applies.

4.2 When is ΔK positive? A bottleneck condition

For finite discrete $\mathcal{X}, \mathcal{Z}, \mathcal{Y}$ under uniform priors over deterministic lookup tables:

$$K(G) \sim |\mathcal{X}| \cdot |\mathcal{Y}|, \quad K(C_{GG}) \sim |\mathcal{X}| \cdot |\mathcal{Z}| + |\mathcal{Z}| \cdot |\mathcal{Y}| + c.$$

So

$$\Delta K = |\mathcal{X}| \cdot |\mathcal{Y}| - |\mathcal{X}| \cdot |\mathcal{Z}| - |\mathcal{Z}| \cdot |\mathcal{Y}| - c.$$

Proposition 6 (Monotonicity and critical bottleneck). *Fix $|\mathcal{X}| = |\mathcal{Y}| = m$ and let $k = |\mathcal{Z}|$. Then $\Delta K(k) = m^2 - 2mk - c$ is strictly decreasing in k , and*

$$\Delta K(k) > 0 \iff k < k^* := \frac{m^2 - c}{2m} \approx \frac{m}{2}.$$

For $c \ll m^2$ and symmetric alphabets, the critical bottleneck is $k^* = m/2$.

Proof. $\partial \Delta K / \partial k = -2m < 0$, so ΔK is strictly decreasing. Setting $\Delta K = 0$ yields $k^* = (m^2 - c)/(2m)$. $\square$

4.3 Three cases, worked out

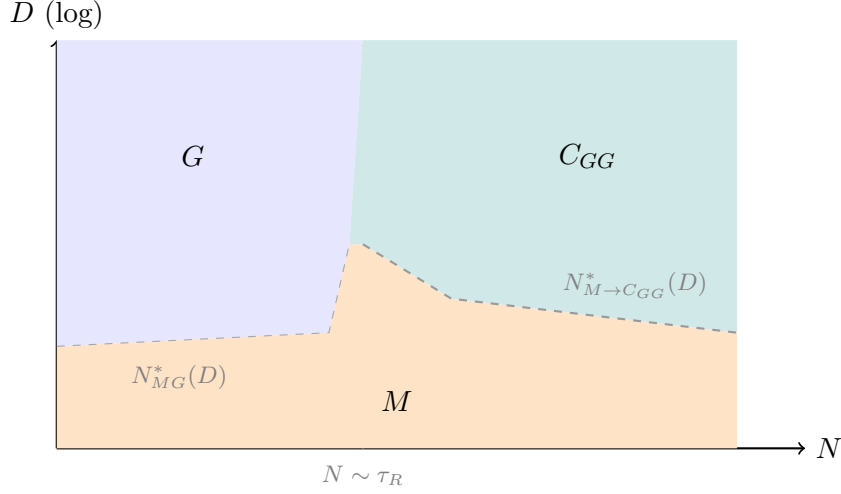
Task family	$K(G)$	$K(C_{GG})$	ΔK
Lookup, $m = 50, k = 5$ (bottleneck)	2500	500	+2000 (large)
Lookup, $m = 50, k = 25$ (marginal)	2500	2500	≈ 0
Lookup, $m = 50, k = 45$ (near-full)	2500	4500	-2000 (large)
Linear-linear, $d = 50$	$d^2 = 2500$	$2d^2 = 5000$	-2500 (K&P regime)
Symmetric Balls & Urns, no bottleneck	$\sim m^2$	$\sim 2m^2$	< 0 (marginal)
Balls & Urns with $k \ll m$ bottleneck	$\sim m^2$	$\sim 2mk$	> 0 (bottleneck)

The framework's central prediction: the sign of ΔK determines whether C_{GG} can emerge, regardless of (N, D) . The linear-linear case sits in the $\Delta K < 0$ column not because linearity itself matters, but because linear maps admit efficient atomic representation — one matrix instead of two.

5 Three Phase Diagrams Indexed by $\text{sign}(\Delta K)$

The framework predicts a distinct phase diagram in each regime.

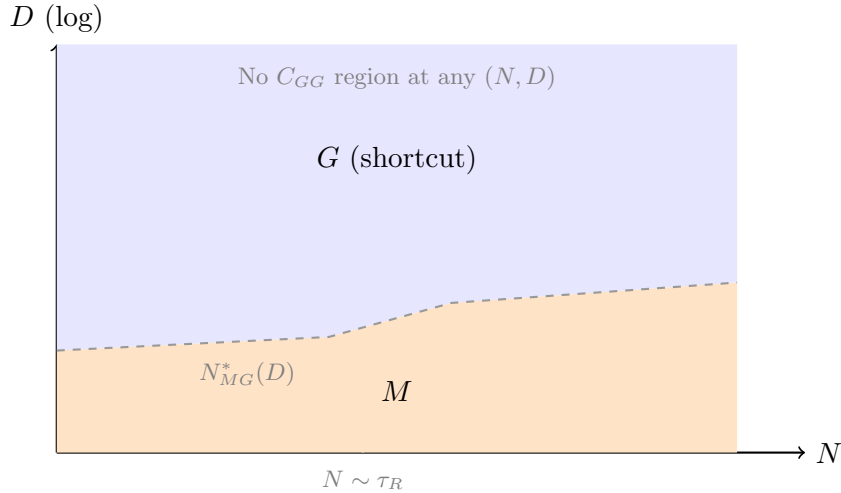
5.1 Regime $\Delta K > 0$: bottleneck tasks, three-region diagram



Regime $\Delta K > 0$: three regions. C_{GG} dominates the post-routing high- D strip.

Pre-routing ($N \ll \tau_R$): Wurgaft's two-region diagram, G dominates at low N and high D , M at high N or low D . Post-routing: C_{GG} replaces G in the high- D region because $K(C_{GG}) < K(G)$ and both have $\Psi = 1$. The right boundary is $N_{M \rightarrow C_{GG}}^*(D)$, which lies strictly to the right of $N_{MG}^*(D)$ because the complexity gap $K_M - K_{C_{GG}} > K_M - K_G$.

5.2 Regime $\Delta K < 0$: shortcut tasks, two-region diagram (Khandelwal–Pavlick)



Regime $\Delta K < 0$: two regions. The atomic shortcut always beats C_{GG} .

The C_{GG} region is empty because $K(G) < K(C_{GG})$ makes the prior always favor G . No amount of training, no amount of task diversity, produces compositional processing. This matches Khandelwal–Pavlick’s observation: the linear-linear composite admits a one-matrix shortcut, and the model uses it.

The only visible dynamics are Wurgaft’s $G \rightarrow M$ transience: at fixed D , the model first uses the shortcut, then memorizes as N grows.

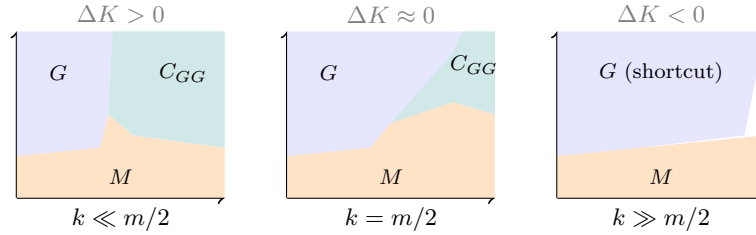
5.3 Regime $\Delta K \approx 0$: marginal tasks, thin C_{GG} region

When ΔK is small (of order the loss-noise scale $|\ell_{C_{GG}} - \ell_G|$), the C_{GG} region exists in principle but is thin, requires large D to appear, and is easily suppressed by finite-sample loss noise. This is the regime where the self-consistency check produced ambiguous results in earlier experiments: the complexity ordering held ($\hat{K}(C_{GG}) < \hat{K}(G)$), but by only a small margin, and the boundary predictions failed to hold at all D .

In practice, this regime is not experimentally useful: predictions are noise-dominated. Push either into $\Delta K > 0$ (by imposing a bottleneck) or into $\Delta K < 0$ (by removing the bottleneck) to get clean signal.

5.4 The morph from three-region to two-region

The framework predicts a specific morphology as ΔK crosses zero. Fix $|\mathcal{X}| = |\mathcal{Y}| = m$ and sweep the bottleneck $k = |\mathcal{Z}|$ from small to large:



The C_{GG} region shrinks and disappears as k crosses $k^* = m/2$. This is a concrete, testable prediction: locate the empirical critical k where the phase diagram flips regime.

6 Phase Boundaries within $\Delta K > 0$

The three boundaries in the $\Delta K > 0$ diagram, obtained by setting $\eta_{Q_1, Q_2} = 0$:

Proposition 7 (Boundaries in the $\Delta K > 0$ regime).

$$N_{MG}^*(D) = \left[\frac{\log 2 \cdot (K_M(D)^\beta - K_G^\beta)}{\gamma(\ell_G - \ell_M)} \right]^{1/(1-\alpha)}, \quad N_{M \rightarrow C_{GG}}^*(D) = \left[\frac{\log 2 \cdot (K_M(D)^\beta - K_{C_{GG}}^\beta)}{\gamma(\ell_{C_{GG}} - \ell_M)} \right]^{1/(1-\alpha)},$$

$$N_{G \rightarrow C_{GG}}^* \approx \tau_R \cdot \log \left[\frac{\gamma \tau_R^{1-\alpha} \Delta_{\text{rout}}}{\log 2 \cdot (K_G^\beta - K_{C_{GG}}^\beta)} \right].$$

Corollary 8 (Composition resists transience). $N_{M \rightarrow C_{GG}}^*(D) > N_{MG}^*(D)$ pointwise when $\Delta K > 0$: the compositional transience boundary sits to the right of Wurgaft’s.

6.1 The full set of D -thresholds inside the compositional strip

Post-routing, each compositional predictor Q has a critical task diversity above which Q beats M : the value D_Q^* at which $\eta_{M,Q}(N, D)$ crosses zero at asymptotic N . Solving $N_{\text{eff}}(\ell_Q - \ell_M) = \log 2 \cdot (K_M(D)^\beta - K_Q^\beta)$ for D gives the ordering:

Proposition 9 (Threshold hierarchy). When $\Delta K > 0$ and stages are symmetric ($k_g \approx k_f =: k$, $K_g^G \approx K_f^G =: K^G$):

$$D_{C_{GG}}^* < D_{C_{MG}}^* \approx D_{C_{GM}}^* < D_{C_{MM}}^* < D_M^*.$$

Each threshold is set by the corresponding complexity gap: lower $K(Q)$ means lower D_Q^* . The lowest-complexity predictor (C_{GG}) appears first as D grows; the highest-complexity nontrivial predictor (C_{MM}) requires the largest D before it survives against M .

Proof. Each threshold is D_Q^* such that $K_M(D_Q^*)^\beta - K_Q^\beta = N_{\text{eff}}(\ell_Q - \ell_M)/\log 2$ at large N . Since $K_M(D)$ is increasing in D , larger K_Q requires larger $K_M(D)$ to compensate, hence larger D_Q^* . The complexity ordering $K(C_{GG}) < K(C_{MG}), K(C_{GM}) < K(C_{MM}) < K_M(D)$ (Proposition 6 and the six-complexity table) gives the threshold ordering. $\square$

6.2 Recombination-driven C_{MM} transient

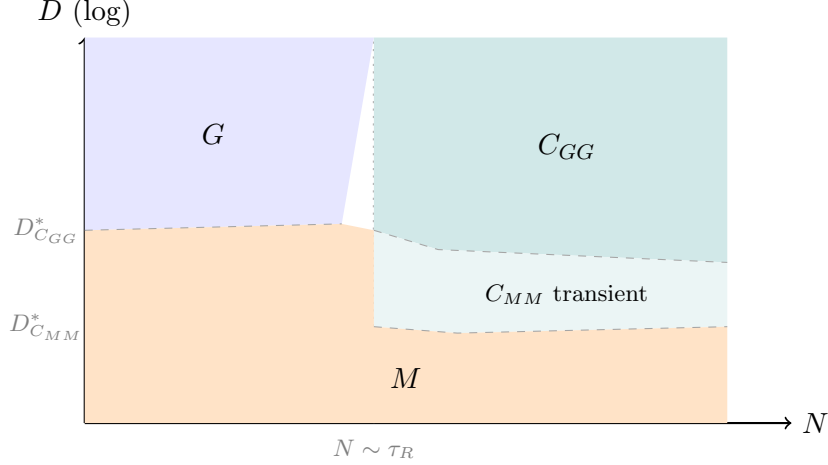
C_{MM} has support over D^2 primitive recombinations, while M ’s support is only the D training pairs. When the training distribution contains recombinations of seen primitives (e.g., the g -only and f -only sequences in the mixed-sequence task allow the model to observe individual primitives across many pairings), C_{MM} handles these correctly while M does not. This gives C_{MM} a likelihood advantage over M on recombination inputs.

Proposition 10 (C_{MM} transient dominance). When training data includes primitive recombinations, and for $D \in (D_{C_{MM}}^*, D_{C_{GG}}^*)$ at moderate N , C_{MM} is the dominant predictor. Below this strip M wins; above it C_{GG} wins on prior.

This creates a transient sub-region for C_{MM} between M and C_{GG} in the phase diagram, at intermediate task diversity.

6.3 Sub-diagram: sub-regions within the $\Delta K > 0$ diagram

Combining Propositions 9 and 10, the symmetric $\Delta K > 0$ diagram admits additional structure at intermediate D :



Symmetric $\Delta K > 0$ diagram with sub-structure. The C_{MM} transient strip requires primitive-recombination structure in training data. Without it, C_{MM} region is absent and only three regions (G, C_{GG}, M) remain.

The asymmetric strategies C_{MG}, C_{GM} do not receive their own regions in the symmetric case: they are dominated pointwise by C_{GG} throughout the compositional strip, per Proposition 9. They appear as their own regions only when stage complexities differ (Section 7).

7 Regime III: Asymmetric Strategies within $\Delta K > 0$

When $\Delta K > 0$ and additionally the stage complexities are asymmetric ($k_g \neq k_f$), the C_{GG} region further partitions.

Proposition 11. C_{MG} dominates C_{GG} iff $D \cdot k_g < K_g^G$ (memorizing g -primitives cheaper than describing their distribution). Mirror: C_{GM} dominates iff $D \cdot k_f < K_f^G$.

Proposition 12 (Asymmetric transience boundaries). *Post-routing, with $K_M(D) - K_{C_{MG}}(D) \approx D \cdot k_f - K_f^G - c$:*

$$N_{M \rightarrow C_{MG}}^*(D) = \left[\frac{\log 2 \cdot (K_M(D)^\beta - K_{C_{MG}}(D)^\beta)}{\gamma(\ell_{C_{MG}} - \ell_M)} \right]^{1/(1-\alpha)} \quad \text{for } Dk_f > K_f^G.$$

The mirror boundary $N_{M \rightarrow C_{GM}}^*(D)$ has k_f replaced by k_g and K_f^G by K_g^G .

When $k_g \ll k_f$: C_{MG} owns a low- D post-routing strip. As D grows past K_g^G/k_g , C_{GG} takes over. The full asymmetric phase diagram has up to four post-routing regions: